DSO 576 Module 6

Individual cleaning submission | national occupation-screening prototype

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Branch URL: https://github.com/emeraldo26/DSO576-Project-/tree/cleantaotianchen123

Question: Which U.S. occupations warrant further human review for AI-transition training when observed AI use and matched task penetration are compared with national employment, outlook, wages, openings and training requirements?

Actual saved results: 3,529 checks passed; 0 failures; 5/5 record checks passed. The master retains 756 occupations; 755 have BLS matches. Sources are unchanged. No commit or push has occurred.

Exact 14-item checklist, items 1-7

Headings follow the supplied assignment, not the earlier provisional interpretation. All items apply; none is Not applicable. Evidence filenames without a path refer to docs/module6_evidence when present, otherwise the 41 files regenerated under outputs/.

# / Exact assignment item	Status	Concrete evidence
1. State the project vision and question	Done	National screening question above; plan section 1 and code/README.md. No score, weights or priority thresholds.
2. Define one row	Done	Source and output grains on page 2; repeated_key_summary.csv. Master: 756 unique SOC occupations.
3. Preserve the source	Unresolved	source_manifest.csv: all six original hashes unchanged. Vendor release/download metadata remains unresolved; preservation itself is Done.
4. Inspect columns and types	Done	conversion_checks.csv: zero unexpected numeric/date failures. Full source_schema.csv and clean_schema.csv are regenerated by the notebook.
5. Standardize text	Done	Derived task keys only lowercase, trim and collapse whitespace. Three punctuation-conflict pairs stay distinct (text_changes.csv and validation_checks.csv).
6. Investigate duplicates	Done	exact_duplicate_summary.csv: 2 groups, 8 involved rows, 6 extras. Repeated keys audited separately. All source rows retained.
7. Handle missing data	Done	before_after_missing.csv: 716 + 321 literal None recoveries; 418 task-type / 587 respondent structural blanks retained with reasons.

Checklist and scope

# / Exact assignment item	Status	Concrete evidence
8. Check suspicious values	Done	suspicious_values.csv and task_vintage_counts.csv flag retained zeros, negative projections and dates. Record 51-2061 keeps published -6.1%.
9. Verify merges	Done	merge_summary.csv: jobs/BLS 755 shared, 1 job-only, 76 BLS-only. Validated join cardinalities; 756 master rows retain Legislators.
10. Reconcile changes	Done	before_after_row_counts.csv and total_reconciliation.csv. All reductions listed on page 4; full view_reductions.csv records excluded-row lineage.
11. Check calculations	Done	metric_denominators.csv; independent Decimal task means for all 756 master SOCs. Unknown tasks excluded, measured zero included. Published units retained.
12. Verify individual records	Done	five_record_checks.csv: 5 PASS. Pre-run expectations in code/module6_record_expectations.json; original, expected and saved actual on pages 5-6.
13. Test reproducibility	Done	reproducibility.json: 41 byte-identical files, sample identical, deliberate source drift rejected. Fresh notebook: 5 code cells, no errors.
14. Document remaining limitations	Done	plan section 9 and limitations below: geography, provenance, task coverage/vintage, SOC-version uncertainty, BLS rounding and wage population.

One row and necessary inputs

job_exposure: one six-digit occupation (756). task_penetration: one supplied task-text observation (17,998). O*NET occupations: one full code including suffix (1,016). O*NET tasks: one occupation-task statement record (18,838). BLS Table 1.2: one coded detail or hierarchy row (1,112). Workbook: original category authority, with Table 1.10 as an independent shared-measure check. All five CSVs and the workbook are needed; unrelated workbook tables are not analytical inputs.

Main measures: observed occupation exposure and mean penetration across unique matched tasks, with total/matched/unmatched task counts. One master row is one job-exposure SOC. A task-view row is one SOC/normalized task. No composite priority score is constructed.

Remaining limitations

California funding conclusions require California employment, projections, wages and program/cost/outcome evidence. Claude use does not measure all AI use or causal job loss. Task coverage and vintages differ; a code match does not prove identical SOC releases. Wages cover a different population from employment, and occupation medians cannot be averaged into a national median.

Upstream metadata remains unresolved: exact Anthropic/O*NET release IDs and original download dates were not supplied; external BLS download provenance is unverified. Local content hashes identify these files, not vendor authenticity. Item 3 remains Unresolved for this metadata only; item 14 is Done because these limitations are documented.

Cleaning decision log | 1 of 2

Implemented, reviewed rules. Raw fields and rows remain intact; canonical companions and explicitly named derived views carry changes. Unexpected source drift, conflicting matches or count/conversion changes raise errors.

Rule	Reason	Check / actual evidence
SCOPE: National screening; compare exposure, task evidence and BLS context	California conclusions need California data; score not approved	No priority score, weights or thresholds emitted
RAW: Immutable hash-pinned inputs; preserve raw fields and rows	Make every change attributable	Source hashes before/after and clean row counts
ID: Text dtype; full IDs retained; derive SOC base separately	Avoid leading-zero loss and suffix erasure	Fullmatch, uniqueness, output text reload; case 1
NONE: Copy code-aligned workbook categories to new canonical fields	CSV blanks erased meaningful no-experience/no-training categories	1037 workbook cell recoveries; other cells equivalent; case 2
MARKER: Em dash to nullable companion and explicit reason	Unavailable or summary not applicable differs from zero/None	Before/after markers; no unexpected conversion failures
STRUCTURAL: Preserve structural blanks as null; retain source reason	Analyst-source records do not supply these metadata	418 type blanks; 587 respondent blanks; source cross-tab
TEXT: Lowercase, collapse whitespace, trim derived keys only	Conservative cross-source task linkage	Three punctuation-conflict regression pairs remain distinct
PEN_DUP: Consolidate only identical values at normalized key; retain lineage	Prevent task lookup fanout and repeated weighting	17998 to 17991; seven exclusions; zero conflicts

Evidence: cleaning_decision_log.csv retains dataset/column, authorization, exceptions, expected reductions and the cleaning-code SHA-256. The two pages reproduce all 16 logged rules without inventing a new scoring policy.

BLS totals and population controls

Measure (thousands)	Published total	831 details	755 matches	Detail - total
employment 2025	170280.8	170267.2	154223.4	-13.6
employment 2035	176198.5	176184.5	158949.6	-14.0
employment change	5917.6	5917.9	4726.9	0.3
annual openings	17461.8	17460.7	15607.2	-1.1

The unmatched 76 BLS details complete the 831-detail totals. Nonzero national residuals are retained, not forced to zero; published rounding/excluded occupational data limit exact reconciliation. Summary and detail rows are never summed together.

Cleaning decision log | 2 of 2

Rule	Reason	Check / actual evidence
ONET_BASE: Select the unique .00 record per base SOC	One description per occupation while preserving suffix detail	1016 to 867; 149 suffix exclusions; all groups have .00
TASK_DUP: One normalized task per six-digit SOC; keep contributing IDs	Do not overweight repeated tasks after suffix collapse	18838 to 18814; 24 exclusions
SUMMARY: Filter Occupation type Line item, never infer from code ending	Avoid hierarchy double counting	1112 to 831; seven legitimate line items ending 0 retained
JOIN: Left join jobs to BLS one-to-one	Keep measured exposure even when BLS is absent	756 master rows; job_only Legislators; case 4
COMPLETE: Optional filter; preserve unmatched master and audit	Explicit population for comparisons requiring BLS	756 to 755; one exclusion
TASK_AVG: Average unique matched task values; report full/ matched denominators	Unmatched is unknown, not measured zero	Independent Decimal mean for every master SOC; case 5
ROUNDED: Preserve published units, zeros, negative projections and percent change	Displayed source precision cannot reproduce all source calculations	285 subtraction residuals <=0.1 thousand; case 3
TOP5: Sort penetration descending then normalized text; retain first five	Stable descriptive task display specified by project	17760 to 3774 with row-level exclusions

Every reduction in a derived view

View	Input	Output	Reduction
penetration_lookup	17998	17991	7
occupation_soc_bridge	1016	867	149
soc_task_bridge	18838	18814	24
matched_task_links	18814	18656	158
bls_line_items	1112	831	281
occupation_master	756	756	0
bls_complete_analysis	756	755	1
occupation_tasks	18656	17760	896
top_five_tasks	17760	3774	13986

All five clean source tables have zero row loss. Reduction reasons are the rules above and before_after_row_counts.csv; row-level evidence is view_reductions.csv. The submitted 50-row sample is 48 master rows plus two BLS controls, deliberately selected for verification, not a representative statistical sample.

Five individual record checks | 1 of 2

Source: five_record_checks.csv. Expected values were authored before execution in module6_record_expectations.json; actual values were reloaded from saved output CSVs. Below are all checked fields and the relevant original fields; full original records remain in the evidence CSV.

Case 1: 00-0000 — PASS

Preserve leading zeros and the national summary control row

Original (relevant source fields): 2025 National Employment Matrix code: 00-0000; Occupation type: Summary; Employment, 2025: 170280.8

Checked field	Expected	Actual saved output
Occupation type	Summary	Summary
2025 employment (000s)	170280.8	170280.8
Is line item	False	False
SOC code	00-0000	00-0000

Case 2: 11-1011 — PASS

Recover the literal None category from the same-code workbook cell, not from an assumption about a blank

Original (relevant source fields): Typical on-the-job training needed to attain competency in the occupation: [empty]; workbook_ojt_original: None

Checked field	Expected	Actual saved output
OJT category	None	None
OJT CSV raw	[empty]	[empty]
OJT missing reason	[empty]	[empty]
OJT workbook raw	None	None

Case 3: 51-2061 — PASS

Keep independently rounded BLS published change and real zeros; em dash is unavailable

Original (relevant source fields): Employment, 2025: 0.2; Employment, 2035: 0.2; Employment change, numeric, 2025–35: 0.0; Employment change, percent, 2025–35: -6.1; Occupational openings, 2025–35 annual average: 0.0; Percent self-employed, 2025: —

Checked field	Expected	Actual saved output
Annual openings (000s)	0.0	0
2025 employment (000s)	0.2	0.2
2035 employment (000s)	0.2	0.2
Published change (%)	-6.1	-6.1
Change (000s)	0.0	0
Self-employed (%)	[empty]	[empty]
Self-employed missing reason	source_unavailable	source_unavailable

Numeric 0 and 0.0 match numerically; IDs/categories match as text. [empty] means the saved empty field, not the literal category None. Valid zeros and published -6.1% are retained.

Five individual record checks | 2 of 2

Case 4: 11-1031 — PASS

Left join retains Legislators and its measured zero exposure

Original (relevant source fields): occ_code: 11-1031; title: Legislators; observed_exposure: 0.0

Checked field	Expected	Actual saved output
BLS match status	job_only	job_only
2025 employment (000s)	[empty]	[empty]
Observed exposure	0.0	0
SOC code	11-1031	11-1031

Case 5: 23-2011 — PASS

Average only the seven matched tasks while explicitly reporting all twelve tasks and five unknown values

Original (relevant source fields): occ_code: 23-2011; title: Paralegals and Legal Assistants; observed_exposure: 0.2927; onet_task_IDs: ['18491', '23994', '21062', '18492', '18493', '1636', '18494', '1639', '1638', '18496', '1642', '1641']

Checked field	Expected	Actual saved output
Matched-task mean	0.133857142857	0.133857142857
Matched unique tasks	7	7
Incomplete coverage	True	True
Task match rate	0.583333333333	0.583333333333
Total unique tasks	12	12
Unmatched unique tasks	5	5

Independent check of the BLS "None" suggestion

The suggestion was treated as a hypothesis, not permission to fill all blanks. The pipeline compared all 1,112 code-aligned workbook rows with the extract, allowing only the documented category differences and verifying numeric equivalence plus shared measures in Table 1.10. The fresh notebook independently read original cells with openpyxl (no cleaning-module import), checked text codes, literal string values, CSV blanks, canonical saved values and exact recovery cell addresses.

Actual independent result: 1,112 coded rows inspected; 716 experience and 321 training cells verified (1,037 total), all PASS. Other missing values were not filled by analogy. This establishes local workbook agreement, not external source authenticity. Evidence: independent_bls_none_check.json and generated bls_category_recovery.csv.

Fresh-kernel execution and handoff

code/module6_cleaning.ipynb completed 5 code cells in a fresh Python 3.11.4 kernel with zero error outputs. It rebuilt sources, displayed validation tables, independently checked categories and reran the 41-file reproducibility/source-drift test. Code, pinned dependencies, evidence and the sample are Git-eligible; raw inputs and bulky outputs remain ignored. Assignment commit ID is pending. See code/README.md for commands.